

Teaching a model to read knee MRIs, when almost nobody labelled them

The goal: for every knee MRI exam, output the probability of 12 findings. The catch: only a handful of exams come with answers. This sheet explains the plan, what we are doing right now, and why.

4,407

training exams (studies),
each with ~4–10 MRI
series

58

have **gold labels**: an
expert's answer for all 12
findings

4,349

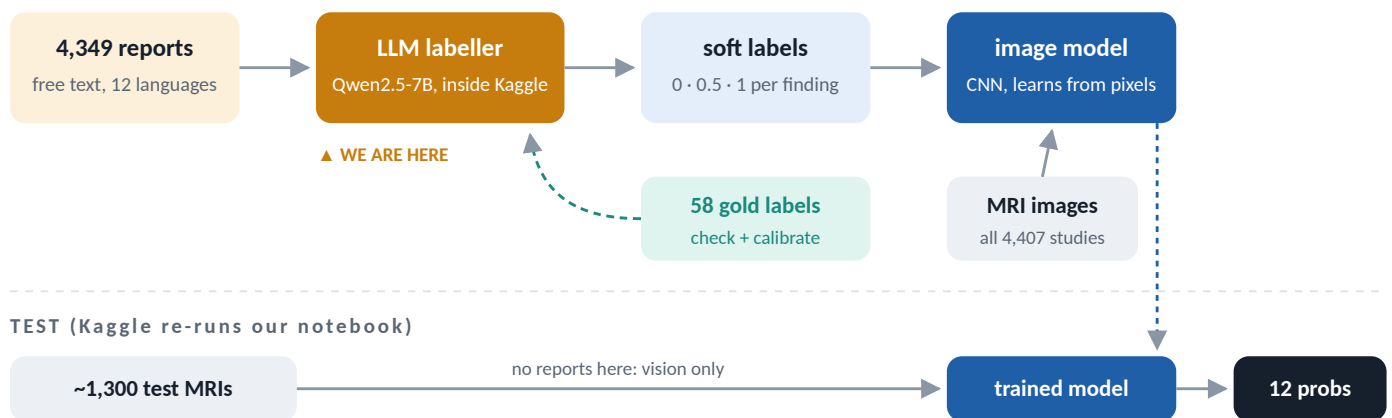
have only a **free-text
radiology report**, in ~12
languages

~1,300

hidden test exams:
images only, no reports,
scored offline within 9 h

The whole plan in one picture

TRAINING (we have reports)



Reports exist only for training. So they are not a model input; they are a **source of labels**. The image model is what gets submitted, and it only ever sees pixels.

How we are scored: macro ROC-AUC

For each finding, AUC asks: *if I pick one exam that has it and one that doesn't, how often does the model give the first a higher score?* 1.0 = always, 0.5 = coin flip. The 12 AUCs are averaged, so the rare findings (fracture, MCL) count as much as the common ones.

Only **ranking** matters, not whether 0.7 "means" 70 %.

Progress so far

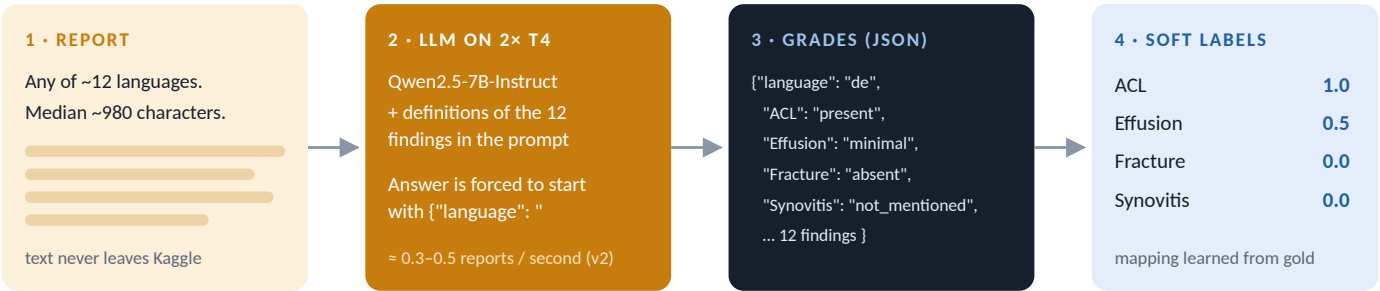
- ✓ **Understand the data:** DICOM, series, 3D geometry, interactive viewer
- ✓ **Dummy submission:** pipeline works offline, score exactly **0.500**
- 3 **Report labeller:** v2 reached AUC 0.82 vs gold; v3 now
- 4 **Image cache + baseline CNN**, then iterate and ensemble

Deadlines: team/entry **15 Oct** • final submission **22 Oct**

The report labeller

A language model reads each radiology report and grades all 12 findings. The 58 gold exams then tell us how those grades translate into labels.

One report, start to finish



Why grades instead of yes / no?

Reports are full of borderline wording: "small effusion", "possible tear", "mild chondral thinning". Whether that counts as 1 is the organizers' decision, not the model's. So the LLM only describes *how* the report says it, and the gold exams decide what it means.

GRADE	WHAT THE REPORT SAYS	DEFAULT LABEL
present	clearly described: "complete ACL tear", "moderate effusion"	1
minimal	small, trace, mild, grade 1, "physiological fluid"	1 (gold may change it)
uncertain	possible, suspected, cannot be excluded	0.5 (gold may change it)
absent	explicitly normal: "ACL intact", "no effusion"	0
not_mentioned	silence: radiologists report positives, so usually normal	0 (gold may change it)

Calibration, in one sentence

For each finding, look at the gold exams that got a borderline grade and set that grade to 0, 0.5 or 1, whichever matches their gold labels best. It happens only if at least 4 gold exams have that grade, so one odd exam can't flip a rule.

Example from v2: "minimal" effusion was gold 1 in 13 cases and gold 0 in 13 → 0.5 is the honest label.

Why a local model, not ChatGPT or Claude?

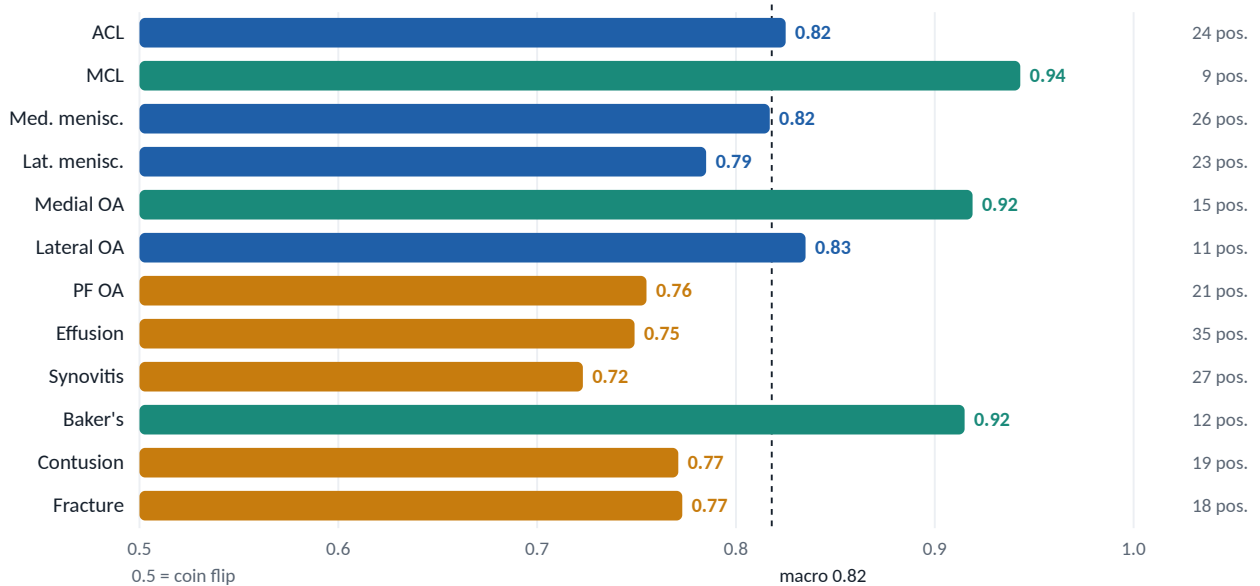
The competition's data-security rule plausibly forbids sending report text to hosted AI services. Running an open-weights model inside the Kaggle notebook keeps every report on Kaggle.

Same reason: share numbers and charts, not report excerpts.

How good are the labels?

Macro AUC **0.82**, mean accuracy **0.81**. Good enough to train on, with clear weak spots.

Agreement with gold, per finding



AUC of the soft labels vs gold (bars) and number of gold positives. With 58 exams, one or two exams flipping moves a bar by several points: read these as rough signals, not exact values.

What goes wrong

too many "present"

ACL • Medial meniscus

10 of 30 ACL "present" are gold 0, same for the medial meniscus. Most likely degeneration or odd signal read as a tear.

v3: only an explicitly stated tear counts.

missed

Synovitis • PF OA

10 gold-positive synovitis exams graded "absent" (gold prevalence 47 %). The organizers label more broadly than reports phrase it.

v3: broader definitions (synovial thickening, chondromalacia).

unreadable

15 % parse errors

The model sometimes adds words ("present (complex tear)"). Those studies were dropped rather than guessed.

v3: tolerant parser + a report of what failed.

Why imperfect labels are still fine

A CNN trained on thousands of mostly-right labels averages out much of the noise. And some disagreement is irreducible: gold may come from experts reading the images, while reports can miss or word things differently.

Next moves

1. Run v3 on gold, compare with v2. 2. Stop polishing, label all 4,349 reports in the background (~2–4 h). 3. Meanwhile build the image cache and a sagittal-only baseline CNN.